

Anti-Forgetting Transfer Learning for Cross-Condition PMSM Stator Fault Diagnosis with Time-Frequency CrossViT

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Abstract—Fault diagnosis models in prognostics and health management (PHM) often face cross-condition distribution shifts, because the same type of motor may operate under different rated capacities or load-related environments. Transfer learning is a natural solution, but direct target-domain fine-tuning can cause catastrophic forgetting and degrade the model’s ability to recognize source-domain faults. This paper studies this problem on a cross-capacity permanent-magnet synchronous motor (PMSM) stator-fault diagnosis task. A time-frequency CrossViT backbone is used to fuse time-domain vibration windows and STFT spectrograms, and an anti-forgetting transfer pipeline is built with teacher-student distillation, LWF-style replay, optional freezing, and EWC regularization. Experiments show that target-oriented adaptation alone improves the new domain but is unstable for source retention, whereas distillation with source replay provides the most reliable trade-off between target adaptation and old-knowledge preservation. In the latest stride-128 setting, the recommended Distill+LWF configuration reaches 91.03% source accuracy and 89.10% target accuracy, and the visualization results further support its anti-forgetting behavior.

Index Terms—PMSM fault diagnosis, cross-condition transfer, time-frequency CrossViT, catastrophic forgetting, knowledge distillation, PHM.

I. INTRODUCTION

Electric motors are key components in manufacturing equipment, transportation systems, and energy conversion devices. Unexpected stator or winding faults may cause efficiency degradation, unplanned downtime, and safety risks. PHM techniques therefore require reliable fault diagnosis models that can identify fault patterns under changing operating conditions. Deep learning has become an effective tool for this purpose because it can learn discriminative representations from high-dimensional time-series measurements and time-frequency images. Recent PHM and industrial monitoring studies further show that deep models and foundation-model-style visual reasoning can improve health assessment and industrial inspection under complex scenarios [1], [2].

In practical motor systems, however, the training and deployment domains are often not identical. A model calibrated on one capacity or load-related condition may face shifted amplitude distributions, frequency components, and time-

frequency structures when it is transferred to another condition. This work therefore describes the task as cross-condition or cross-capacity fault diagnosis rather than cross-capacity diagnosis, avoiding confusion with the signal-processing term cross-capacity spectrum. Existing transfer diagnosis and domain adaptation methods attempt to reduce such distribution discrepancy and improve target-domain recognition [3]–[7].

A limitation of many transfer diagnosis studies is that target-domain accuracy is treated as the dominant goal, while source-domain knowledge retention is rarely evaluated. In industrial PHM, a diagnostic system may be reused across multiple motors or repeatedly switch among working conditions. If direct fine-tuning adapts the model to a new target domain but destroys the source decision boundary, the resulting model becomes unreliable for multi-condition deployment. This is a catastrophic forgetting problem in a transfer-learning context.

To address this issue, this paper formulates cross-capacity PMSM stator fault diagnosis as a dual-objective transfer problem: the model should adapt to the target capacity while preserving source-domain fault knowledge. Methodologically, the proposed pipeline combines a time-frequency CrossViT backbone with anti-forgetting strategies from continual learning. The backbone extracts local temporal patterns from the vibration branch, spectral and time-frequency structures from the spectrogram branch, and complementary diagnostic cues through attention-based multi-modal fusion. The transfer stage is designed as an anti-forgetting pipeline: a source-domain teacher constrains the student with softened decision distributions, replayed source windows stabilize old decision boundaries, and optional freezing or EWC limits destructive parameter drift.

The main methodological contributions are summarized as follows. First, we construct a time-frequency CrossViT diagnostic model tailored to cross-condition PMSM stator-fault diagnosis, so that waveform-level transients and spectrogram-level fault signatures can be represented in a unified framework. Second, we design a transfer pipeline with explicit anti-forgetting ability, in which teacher-student distillation and LWF-style replay jointly maintain source-domain decision

structure during target adaptation. Beyond these two method components, the paper reports a systematic ablation from basic fine-tuning to domain adaptation and continual-learning variants, demonstrating when target adaptation and source retention become compatible or conflicting.

II. RELATED WORK

A. Deep Learning and Multi-Modal Fault Diagnosis

Deep learning-based fault diagnosis has been widely studied for rotating machinery. CNN-based and transformer-based models can learn local and long-range signal structures in an end-to-end manner. Time-frequency representations, such as STFT and wavelet-based maps, are also frequently used because they reveal frequency and energy distributions that may not be obvious in raw time signals. Recent work has further combined convolutional modules and attention mechanisms to extract local and global representations for bearing diagnosis [11]. Dual-domain signal fusion methods also indicate that time-domain and frequency-domain signals provide complementary information for cross-domain diagnosis [6].

B. Transfer Learning and Domain Adaptation

Transfer learning uses source-domain knowledge to improve target-domain learning. In fault diagnosis, the source domain usually contains sufficient labeled samples, while the target domain may contain limited or unlabeled samples under another machine, load, or capacity condition. Domain adaptation methods reduce feature discrepancy between domains through distribution matching or adversarial learning. DSCNN and CAN minimize MK-MMD to learn transferable features [3], [4], [12]. DANN introduces a domain discriminator and gradient reversal layer to learn domain-invariant representations [13]. Other methods jointly align marginal and conditional distributions, or use Wasserstein distance and adversarial training to improve transfer performance [11], [14], [15].

Although these methods are effective for target-domain adaptation, they do not always preserve source-domain performance. Distribution alignment may push the representation toward domain invariance, but domain invariance alone cannot guarantee old decision-boundary retention. This motivates the use of continual-learning strategies in cross-condition diagnosis.

C. Continual Learning and Knowledge Retention

Continual learning studies how a model can learn new tasks without forgetting old ones. EWC protects important parameters by using a Fisher-information-weighted penalty [8]. Knowledge distillation transfers the softened output distribution of a teacher model to a student model [9]. LWF constrains the new model to preserve old-task responses during new-task learning [10]. In this paper, these strategies are adapted to cross-capacity fault diagnosis. The source-domain model is regarded as a teacher, and source replay is used to maintain source decision boundaries while learning target-domain features.

III. METHOD

A. Problem Definition and Multi-Modal Construction

Let $D_s = \{(x_i^s, y_i^s)\}_{i=1}^{n_s}$ denote the labeled source domain collected from a 1.0 kW PMSM and $D_t = \{(x_i^t, y_i^t)\}_{i=1}^{n_t}$ denote the target domain collected from a 3.0 kW PMSM. The two domains share the same stator-fault label space, but their feature distributions differ due to motor-capacity variation, i.e., $P_s(x) \neq P_t(x)$. Each processed sample is represented as a three-channel time-series window. The window is also transformed into a time-frequency spectrogram by STFT. Thus, the input of each sample contains two complementary modalities: a time-domain signal x_{time} and a spectrogram x_{spec} .

B. Multi-Modal CrossViT Backbone

The backbone contains a time-domain encoder, a spectrogram encoder, a cross-modal attention fusion module, and a classifier head. The time-domain encoder projects the segmented waveform into tokens and models local temporal dependencies caused by impacts and transient variations. The spectrogram encoder uses two-dimensional patch embedding and transformer blocks to capture frequency bands and time-frequency energy structures. The two encoded representations are then fused at the modality level so that temporal and spectral cues jointly determine the fault category:

$$F_t = E_t(x_{time}), \quad F_s = E_s(x_{spec}), \quad (1)$$

$$F_{fusion} = A_{cross}(F_t, F_s), \quad \hat{y} = C(F_{fusion}), \quad (2)$$

where E_t and E_s are the two encoders, $A_{cross}(\cdot)$ denotes cross-modal attention fusion, and $C(\cdot)$ is the classifier. The purpose of this architecture is not only to improve recognition accuracy, but also to provide a stable feature basis for subsequent anti-forgetting transfer.

C. Source-Domain Pre-Training

The first stage trains the multi-modal CrossViT on the source domain. The source loss is the standard cross-entropy loss:

$$\mathcal{L}_{src} = -\frac{1}{B} \sum_{i=1}^B \sum_{c=1}^C \mathbf{1}(y_i = c) \log p_i(c). \quad (3)$$

The trained model is saved as a teacher model f_{θ^*} and also provides initialization for target adaptation.

D. Anti-Forgetting Transfer With Distillation, LWF and Freezing

During target adaptation, direct fine-tuning can overwrite source-domain representations. To reduce forgetting, the proposed strategy combines three mechanisms. First, layer freezing is optionally used to reduce large parameter drift in early encoders. Second, a teacher model provides soft labels. Third, source replay following the LWF idea mixes a small ratio of source-domain samples with target-domain samples.

The knowledge distillation loss is

$$\mathcal{L}_{KD} = T^2 \text{KL}(\sigma(z_s/T), \sigma(z_T/T)), \quad (4)$$

Algorithm 1: Multi-Modal Source-Domain Pre-Training

Input: source domain D_s , model f_θ , maximum epochs E_s .
Output: teacher model f_{θ^*} and initialization parameters.

- 1: Initialize time encoder, spectrogram encoder, fusion module and classifier.
- 2: **for** epoch = 1, ..., E_s **do**
- 3: Sample a mini-batch $(x_{time}^s, x_{spec}^s, y^s)$ from D_s .
- 4: Compute $F_t = E_t(x_{time}^s)$ and $F_s = E_s(x_{spec}^s)$.
- 5: Fuse features by $F_{fusion} = \mathcal{A}_{cross}(F_t, F_s)$.
- 6: Predict $\hat{y}^s = C(F_{fusion})$ and compute $\mathcal{L}_{src}$.
- 7: Update model parameters by back-propagation.
- 8: **end for**
- 9: Save the best source-domain model as teacher f_{θ^*} .

Fig. 1: Source-domain pre-training for stable fault knowledge learning.

Algorithm 2: Anti-Forgetting Transfer With Freezing, EWC, Distillation and LWF

Input: teacher f_{θ^*} , source replay set R_s , target set D_t , temperature T , weight α .
Output: adapted model $f_{\theta'}$.

- 1: Initialize student $f_\theta \leftarrow f_{\theta^*}$.
- 2: Freeze selected embedding or encoder layers if required.
- 3: Estimate Fisher matrix on D_s when EWC is enabled.
- 4: **for** epoch = 1, ..., E_t **do**
- 5: Sample target mini-batch B_t and replay mini-batch $B_s \subset R_s$.
- 6: Compute student logits z_s and teacher logits z_T .
- 7: Compute hard-label loss $\mathcal{L}_{CE}$ on available labels.
- 8: Compute distillation loss $\mathcal{L}_{KD}$ from softened teacher outputs.
- 9: Add optional $\mathcal{L}_{EWC}$ to protect important parameters.
- 10: Optimize $\mathcal{L} = (1 - \alpha)\mathcal{L}_{CE} + \alpha\mathcal{L}_{KD} + \lambda_e\mathcal{L}_{EWC}$.
- 11: **end for**
- 12: Return the model with the best Balance Score.

Fig. 2: Anti-forgetting transfer procedure used in the final recommended pipeline.

where z_s and z_T are student and teacher logits, T is the temperature, and $\sigma(\cdot)$ is softmax. The final loss is

$$\mathcal{L} = (1 - \alpha)\mathcal{L}_{CE} + \alpha\mathcal{L}_{KD} + \lambda_e\mathcal{L}_{EWC}, \quad (5)$$

where α controls the distillation weight and $\mathcal{L}_{EWC}$ is optional.

E. Domain Adaptation Candidates and Model Selection

DANN, MMD and pseudo-labeling are treated as ablation candidates. DANN uses a domain discriminator connected by a gradient reversal layer. MMD minimizes feature-distribution discrepancy. Pseudo-labeling selects high-confidence target predictions as temporary labels. Since these methods may improve target-domain accuracy while damaging source retention, the final model is selected by a dual-domain Balance Score:

$$S_{bal} = A_t - \beta \max(0, A_s^{ref} - A_s), \quad (6)$$

where A_t is target accuracy, A_s is source accuracy, and A_s^{ref} is a source retention reference. A higher S_{bal} indicates a better balance.

Algorithm 3: Ablation Candidates and Dual-Domain Model Selection

Input: candidate set $\mathcal{M}$, source test set, target test set.
Output: selected model m^* .

- 1: Construct candidates: fine-tuning, freezing, EWC, Distill, Distill+LWF, DANN, MMD and pseudo-labeling.
- 2: **for each** candidate $m \in \mathcal{M}$ **do**
- 3: Train or adapt m under the same source-target task.
- 4: Evaluate source retention $A_s(m)$ and target accuracy $A_t(m)$.
- 5: Compute Balance Score $S_{bal}(m)$.
- 6: **end for**
- 7: Select $m^* = \arg \max_m S_{bal}(m)$.
- 8: Generate confusion matrices, training curves and attention maps for m^* .

Fig. 3: Ablation-driven model selection for dual-objective transfer diagnosis.

IV. EXPERIMENTS

A. Dataset and Experimental Settings

The experiments use Dataset 2 in the collected dataset document, namely the KAIST/Mendeley Data PMSM stator-fault dataset [16]. The public benchmark was collected on a PMSM fault-diagnosis testbed containing three-phase motors with capacities of 1.0 kW, 1.5 kW, and 3.0 kW. The motors were tested at 3000 RPM under the same load setting. The measurement system used an IEPE piezoelectric accelerometer for vibration and three current transformers for phase-current acquisition, connected to NI data-acquisition modules. Each condition was recorded for 120 s; vibration signals were sampled at 25.6 kHz and current signals at 100 kHz. The artificial stator faults include inter-turn and inter-coil short circuits with multiple severities, making the dataset suitable for evaluating capacity-related distribution shifts.

In this paper, the 1.0 kW motor is used as the source domain and the 3.0 kW motor is used as the target domain. The released data are processed into three-channel time-series windows with length 1024 and stride 128, and each window is further transformed into a $3 \times 128 \times 128$ spectrogram by STFT. The source and target domains share a 16-class diagnosis protocol constructed from the available normal and stator-fault conditions. All compared methods use the same preprocessing, data split, backbone, and evaluation pipeline to ensure that the reported differences mainly come from the transfer strategies.

B. Overall Quantitative Results

Table II summarizes the complete experimental evolution. The observation column has been removed to keep the configuration field readable. The results include the original stride-256 series and the latest stride-128 series. EXP1 gives the single-domain baseline. EXP2 verifies that direct cross-capacity generalization fails. EXP3 V1 to V4 then gradually introduce fine-tuning, model scaling, freezing, continual-learning strategies, and domain adaptation candidates.

The progression in Table II shows that a naive cross-capacity model is not reliable. EXP2 obtains high source-domain accuracy but almost fails in the target domain. V1 confirms

TABLE I: Dataset Settings for Cross-Capacity PMSM Fault Diagnosis

Item	Description
Public source	KAIST/Mendeley PMSM stator-fault dataset
Motor capacities	1.0, 1.5 and 3.0 kW; this work uses 1.0 kW $\rightarrow$ 3.0 kW
Operating condition	3000 RPM under the same torque-load condition
Original sensors	IEPE accelerometer and three current transformers with NI DAQ
Original sampling	25.6 kHz vibration, 100 kHz current, 120 s per condition
Processed input	1024×3 time-series window and $3 \times 128 \times 128$ spectrogram
Fault protocol	16-class stator-fault diagnosis setting

that basic fine-tuning is not sufficient. V2 improves target-domain performance through model scaling and freezing, yet source-domain retention remains very low, indicating severe catastrophic forgetting.

The V3 and V4 results further show the difference between target-oriented adaptation and anti-forgetting adaptation. Direct fine-tuning and domain-alignment candidates such as DANN and MMD can raise target-domain accuracy, but they often damage the source-domain decision boundary. EWC and simple freezing are more conservative, but they restrict target-domain adaptation and therefore do not provide the best overall trade-off.

The strongest finding is that knowledge distillation combined with LWF-style source replay gives the most stable dual-domain behavior. In the latest stride-128 setting, Distill+LWF with $T = 2$ and $\alpha = 0.3$ achieves 91.03% source accuracy, 89.10% target accuracy, and a Balance Score of 99.10. The latest V4 distillation setting with $T = 1.5$ and $\alpha = 0.25$ obtains the highest source retention, 94.23%, while still maintaining 87.18% target accuracy. Therefore, Table II is sufficient to support the main quantitative conclusion, and detailed ablation results are discussed in text rather than using another dense table.

More specifically, freezing only early layers protects a small part of the representation but restricts the feature shift required by the target capacity condition. EWC introduces a Fisher-weighted constraint on important parameters, yet its performance is sensitive to the regularization weight and does not consistently prevent forgetting. DANN and MMD are useful as target-oriented domain adaptation candidates, but their low source-domain retention indicates that domain invariance alone is not equivalent to continual reliability. Pseudo-label learning is the least stable strategy in this task because the initial target predictions are not confident enough under the large capacity-induced distribution gap.

For this reason, the paper treats DANN, MMD and pseudo-label learning as ablation candidates rather than the final recommended solution. The final recommendation is a distillation-and-replay strategy: the teacher model preserves the source-domain soft decision structure, while replayed source windows prevent the classifier head from drifting

completely toward the target domain.

C. Visualization Analysis

The following figures are placed after the quantitative discussion rather than immediately after the tables. This separates numerical comparison from visual evidence. The confusion matrices highlight class-level source retention and target adaptation, while the training curves show the convergence behavior of the selected distillation configurations. In addition, an encoder self-attention map is visualized for a representative correctly classified sample. Since the implemented cross-modal fusion operates after global feature pooling, post-pooling cross-attention weights describe modality-level fusion rather than token-level localization. Therefore, the attention analysis focuses on the self-attention maps inside the time-domain and spectrogram encoders, where token-level temporal and spectral dependencies are explicitly modeled.

Fig. 4 corresponds to the latest V4 configuration. The two matrices show that the model keeps a clear diagonal structure in both domains, which means the distillation objective does not simply overfit the target domain.

Fig. 5 corresponds to the latest V3 Distill+LWF configuration, which obtains the highest Balance Score. Compared with target-only adaptation, the source-domain confusion matrix remains much more diagonal, demonstrating the effect of replay and teacher constraints. The target-domain matrix also keeps most samples around the diagonal, indicating that anti-forgetting does not simply sacrifice adaptation for retention.

Fig. 6 shows stable convergence during both source pre-training and target fine-tuning. The gap between training and validation accuracy is moderate in the distillation-based settings, indicating that the teacher constraint and replay samples help stabilize adaptation.

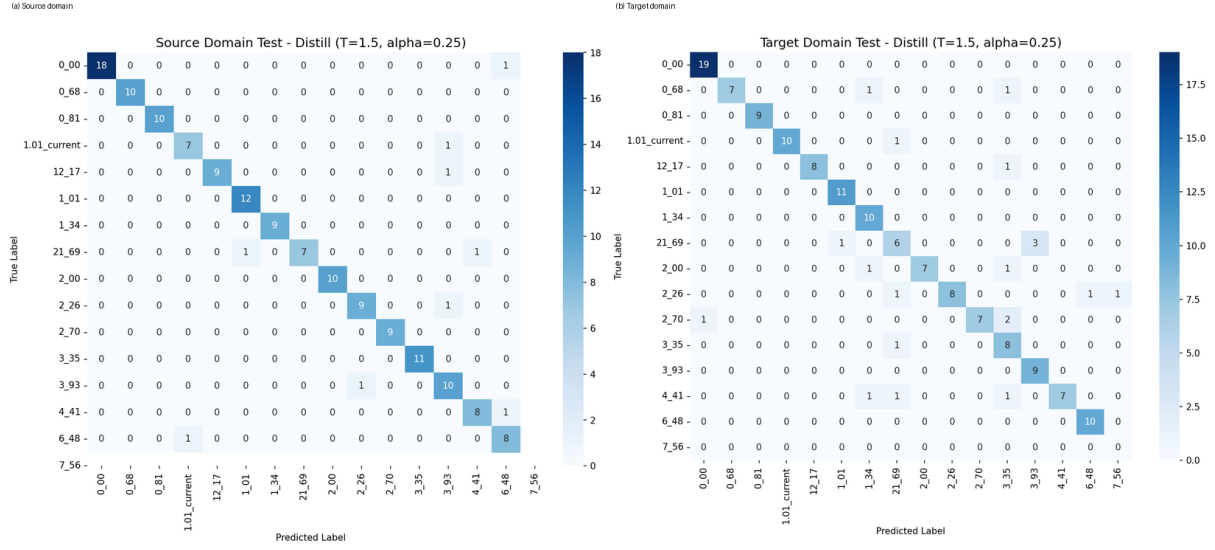
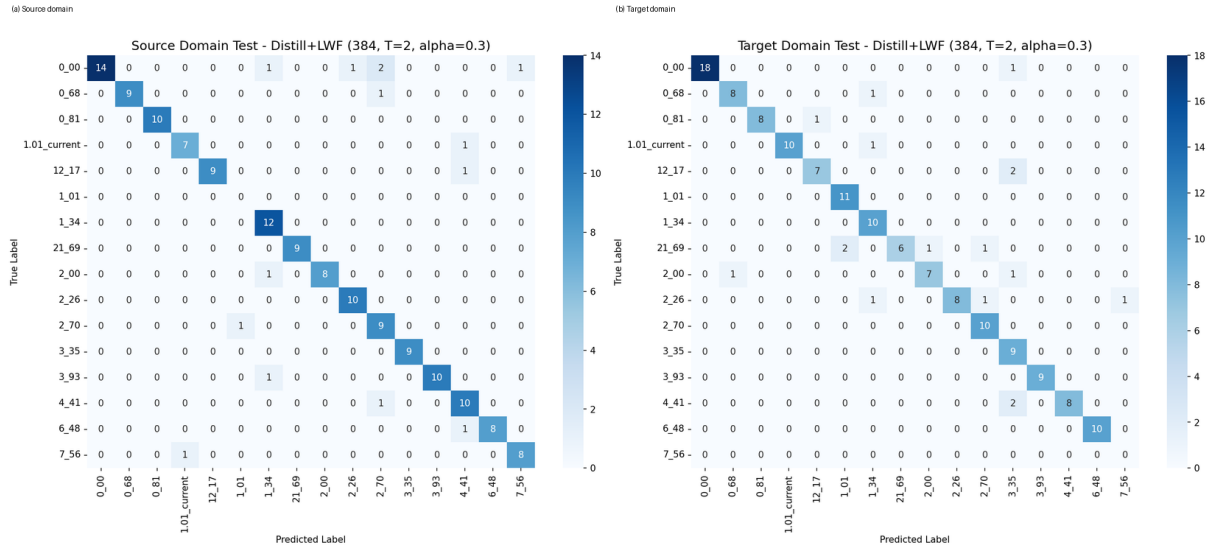
Fig. 7 shows a high-confidence correctly classified target-domain sample. The time-domain attention heads contain localized and banded structures, indicating that the model captures dependencies among neighboring signal patches. The spectrogram attention heads exhibit structured vertical or block-like responses over selected spectral tokens, suggesting that the frequency branch emphasizes discriminative time-frequency regions rather than uniformly attending to the whole spectrogram.

V. CONCLUSION

This paper studies cross-capacity PMSM stator fault diagnosis from the perspective of anti-forgetting transfer learning. Instead of optimizing target-domain accuracy alone, the task is formulated as a dual-objective problem that jointly evaluates target adaptation and source-domain retention. The proposed pipeline combines a time-frequency CrossViT backbone with layer freezing, EWC, teacher-student distillation, and LWF-style source replay. Experiments from EXP1 to EXP3 V4 show that pure fine-tuning and target-oriented domain adaptation candidates can improve target performance but often cause severe source-domain forgetting. In contrast, the Distill+LWF

TABLE II: Evolution of EXP1, EXP2, EXP3 V1-V4 and Latest Stride-128 Results

Stage	Setting / Best configuration	Source (%)	Target (%)	Balance
EXP1	1.0 kW single-domain baseline	87.82 / 90.00	—	—
EXP2	1.0 kW $\rightarrow$ 3.0 kW, no adaptation	94.23 / 85.00	4.49 / 10.00	—
EXP3 V1	Basic fine-tuning	12.50	27.50	27.50
EXP3 V2	Model scaling + freezing	5.00	67.50	67.50
EXP3 V3	Distill+LWF, $T = 2$, $\alpha = 0.3$	77.50	65.00	75.00
EXP3 V4	Distill, $T = 1.8$, $\alpha = 0.3$	60.00	70.00	80.00
Latest V3	Distill+LWF, $T = 2$, $\alpha = 0.3$	91.03	89.10	99.10
Latest V4	Distill, $T = 1.5$, $\alpha = 0.25$	94.23	87.18	97.18

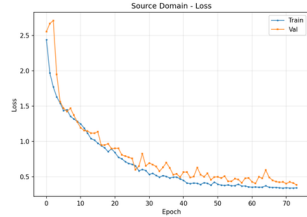
Fig. 4: Source-domain and target-domain confusion matrices of Distill ($T = 1.5$, $\alpha = 0.25$).Fig. 5: Source-domain and target-domain confusion matrices of Distill+LWF ($T = 2$, $\alpha = 0.3$).

strategy provides the most stable balance. In the latest stride-128 setting, it achieves 91.03% source accuracy, 89.10% target accuracy, and a Balance Score of 99.10.

Future work will extend the current two-domain setting to multi-capacity and multi-source continual transfer, where

the model must incrementally absorb several motor domains without repeated full retraining. Another direction is to reduce dependence on target labels through semi-supervised, open-set, or test-time adaptation, while maintaining explicit source-

(a) Distill T=1.5, alpha=0.25



(b) Distill+LWF T=2, alpha=0.3

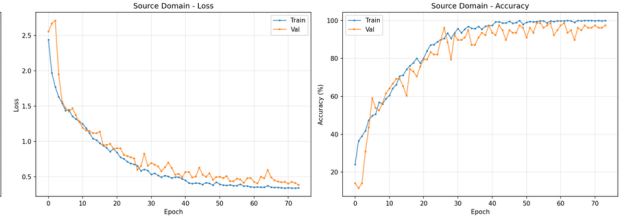


Fig. 6: Training curves of representative distillation-based configurations.

Encoder Self-Attention Visualization (Time-Domain & Spectrogram)

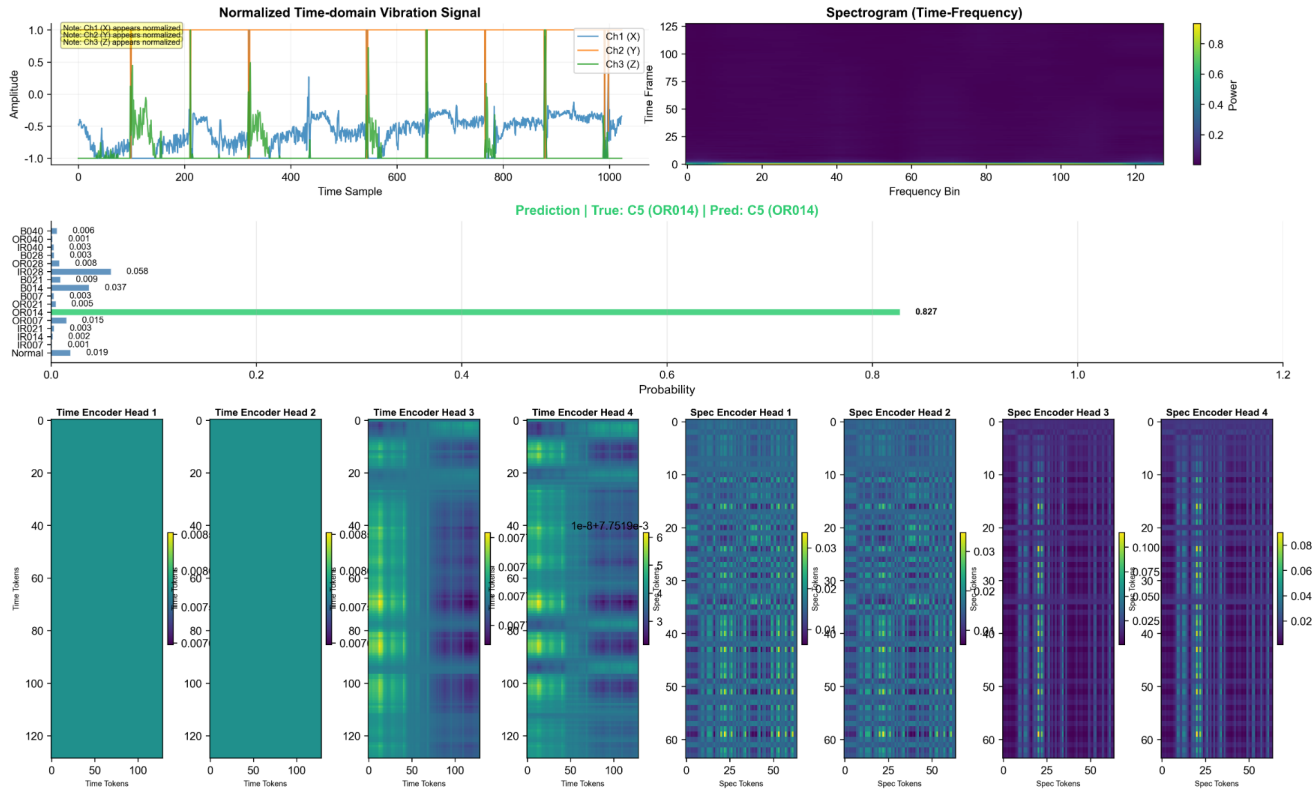


Fig. 7: Encoder self-attention visualization for a correctly classified C5 sample. The upper part shows the normalized three-channel vibration signal, spectrogram and prediction probability, while the lower part visualizes representative self-attention heads from the time-domain and spectrogram encoders.

domain retention constraints. More practical deployment also requires online change detection, uncertainty estimation, and memory-budget control, because PHM systems may face unknown faults and limited replay storage. Finally, combining the proposed anti-forgetting pipeline with physics-informed motor indicators may further improve reliability and interpretability for real-time industrial deployment.

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